NANYANG JUNIOR COLLEGE
Year 2 Preliminary Examination

H2 GEOGRAPHY 9173/02

Paper 2 21 Sept 2026

3 hours

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

Answer Question 1 in **Section A.**
Answer Question 2 in **Section B**.
Answer **one** question in **Section C**.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer Questions 1 and 2 and
EITHER Question 3 **OR** Question 4

Section A

Cluster 4 Fieldwork

1 A group of 10 students wanted to investigate how the business community in Singapore is responding to climate change. They selected an **"**Eco-Commercial District" in Jurong East, which contains a mixture of transnational corporations (TNCs) and local small-and-medium enterprises (SMEs). The investigation focused on the businesses' climate change mitigation measures and their willingness and capacity to invest in such measures.

They developed the following hypothesis to guide their investigation:

'Large transnational corporations (TNCs) exhibit a higher willingness and capacity to adopt climate change mitigation measures than local small-and-medium enterprises (SMEs).'

The students divided themselves into two groups and conducted their fieldwork on a Tuesday in October, from 2.00 pm to 5.00 pm. They moved around the commercial district and approached business managers or shop tenants who were willing to participate in the survey. In total, they surveyed six business representatives: two TNC managers and four SME representatives.

In addition to the questionnaire survey, the students conducted physical audits of all six-business premises surveyed—two TNCs and four SMEs. They recorded visible features associated with climate change mitigation, such as energy-efficient lighting, environmentally friendly packaging and energy-efficient appliances. They also took photographs of green initiatives adopted by the TNCs and SMEs, such as recycling facilities and other environmentally sustainable practices.

Resource 1 shows photographs taken by the students of business premises and green initiatives in the commercial district.

Resource 2 shows the questionnaire administered to the TNC and SME representatives.

Resource 3 shows selected data collected from the questionnaire survey.

Resource 4 shows a comparison of the physical audit of sustainability performance for the two TNCs and four SMEs.

(a) Explain **one** strength and **one** weakness of the students' hypothesis. [4]

(b) Explain **three** potential safety risks that the students may encounter while conducting fieldwork in the commercial district and how each risk could be minimised. [6]

(c) Using Resource 1, describe two research ethics issues that may arise during the investigation and suggest how each could be addressed. [4]

(d) With reference to Resource 2, explain how the design of the survey questionnaire used to collect data from the business community could be improved. [6]

(e) Explain why it would be useful to conduct a short interview after administering the questionnaire. [4]

(f) With reference to Resource 4, explain the strengths and limitations of using a radar chart to represent the audit scores for the various organisations. [6]

(g) With reference to all the Resources, evaluate the validity of the students' investigation of how the business community in Singapore is responding to climate change. [10]

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 5 shows a choropleth map of the global distribution of ecological footprint per person (in global hectares). Resource 6 shows a table of Municipal Solid Waste (MSW) generation, recycling, and disposal by key material categories for the United States (based on 2018 EPA data). Resource 7 is an excerpt from the United Nations report "The Weight of Cities: Resource Requirements of Future Urbanisation". Resource 8 is an aerial photograph showing the scale of the Bantar Gebang landfill in Indonesia and its proximity to surrounding infrastructure and waterways.

(a) Describe the global spatial distribution of ecological footprint per person shown in Resource 5. [4]

(b) Account for the variations in the global spatial distribution of ecological footprint per person shown in Resource 5. [6]

(c) With reference to Resource 6, describe the patterns of municipal solid waste generation, recycling, and disposal across the different material categories. [5]

(d) With reference to Resource 7, explain how projected urban trends drive the escalation of material consumption by cities. [4]

(e) Using Resource 8, explain the environmental hazards and liveability risks associated with the accumulation of solid waste in urban peripheries. [5]

(f) Explain why reducing an urban area’s ecological footprint presents trade-offs between the economic and environmental dimensions of sustainable development. [6]

[Turn over

Section C

Answer EITHER question 3 OR question 4.

3 Evaluate the extent to which urban strategies address the root causes of elderly vulnerability rather than merely managing their symptoms. [20]

4 'The success of climate change responses depends more on partnership between the state, firms and non-governmental organisations (NGOs) than on the individual actions of any single actor.' Evaluate this statement. [20]

End of Paper